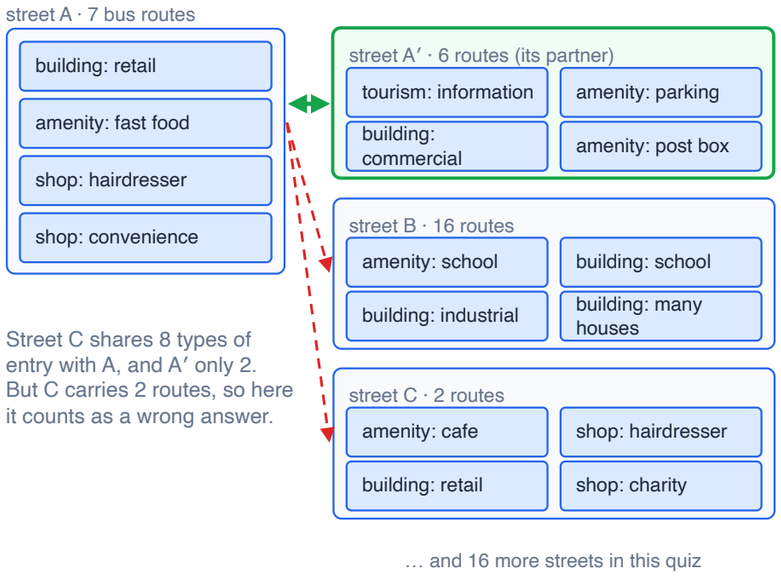
Stage 3: how CityFM matches streets by their number of bus routes, and why it learns little

A street is described by the things near it, read as types without names. Bus routes decide only which streets are paired.

its partner: about the same number of bus routes made more similar: its partner made less similar: the other streets in the quiz (about 18)

1 Training

Each street on a bus route is paired with another that carries about the same number of routes, anywhere in Leeds. The model is adjusted so that each street's context is more similar to its partner's than to the other streets in the quiz. Repeated for 270 quizzes.



Street C shares 8 types of entry with A, and A′ only 2. But C carries 2 routes, so here it counts as a wrong answer.

… and 16 more streets in this quiz

What is trained
BERT and the text projection, the same text side as stage 1, at a tenth of its learning rate. The footprint side is not changed, but the text embeddings it was trained to match are.

2 How a quiz is drawn

All segments are shuffled and cut into windows of 1,024. In each window, the segments on a bus route are taken in turn and paired with the first unpaired one within one route of them. No pair starts from a number of routes already used in the quiz.

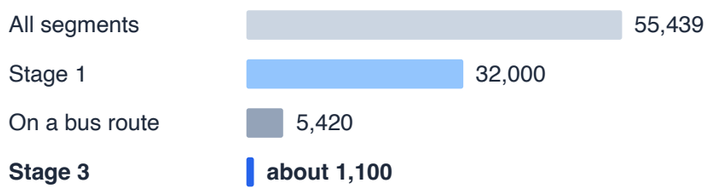
1,024 segments in a window → **about 100** on a bus route → **about 10 pairs** (6 to 15)

The 10 pairs of the window in panel 1, as bus routes on each side

7 – 6	1 – 1	13 – 14	5 – 5	3 – 4
8 – 7	2 – 1	12 – 11	9 – 8	16 – 16

The first pair is streets A and A′. Across many windows, partners are a median of 6.6 km apart, and 93% are more than 1 km apart.

Road segments used in one epoch

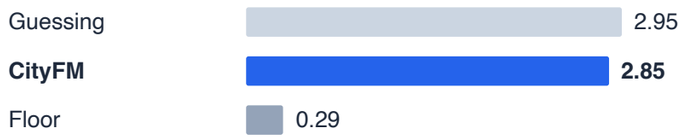


Stage 3 uses 2% of the segments in an epoch, against 58% for stage 1. Five epochs give 270 quizzes.

3 Result · the training loss

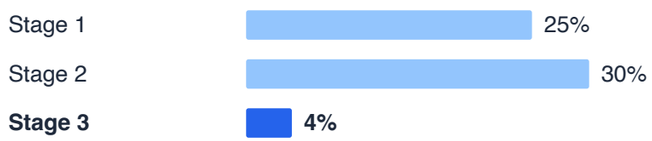
There is no separate test for stage 3. The loss is low when each street's context is more similar to its partner's than to the decoys. With about 10 pairs per quiz, guessing gives 2.95.

Stage 3 loss, last 25 steps (lower is better)



The loss stays close to guessing for all 270 steps.

Share of the way from guessing to the floor



Stages 1 and 2 add 4 quizzes of 64 pairs per step, so their losses are on another scale. The share of the way can be compared across stages.

What it shows. Stage 3 barely learns. Its loss ends 4% of the way from guessing to the floor, against 25% and 30% for stages 1 and 2. Pairing streets by their number of bus routes gives the model little to learn, and each epoch uses only about 1,100 of the 55,439 segments. **Next:** replace the target with each street's network profile (angular integration and choice at several radii), which every segment has.

Guessing: every candidate scored the same, log(2n – 1) for n pairs. **Floor**: partners identical and decoys opposite, which cannot be reached but shows the scale. Panels 1 and 2 show one real Leeds window, with 4 of the commonest entries in each context.